

(b) Resistor R has resistance $45\ \Omega$.

Assume that there is no p.d. across the diode when it is conducting.

(i) Determine the peak power P_0 in the resistor.

$$P_0 = \dots\dots\dots \text{ W [2]}$$

(ii) On Fig. 8.3, sketch the variation of the power P in the resistor with t between $t = 0$ and $t = 2T$.

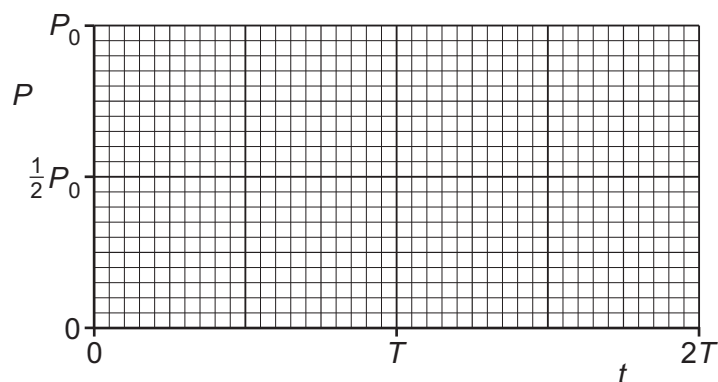


Fig. 8.3

[3]

(iii) Use the answer in (b)(ii) to explain why the mean power in the resistor is $\frac{1}{4}P_0$.

.....

 [2]

(iv) Use the information in (b)(iii) to determine the r.m.s. value of V_{OUT} .

$$\text{r.m.s. voltage} = \dots\dots\dots \text{ V [1]}$$

[Total: 10]

